Sorry to be the bearer of bad news, but learning Proofs is **not** an April Fool's Prank

Predicate Logic Proofs

01 April 2026

Please remember …

The new rules we added simply allow us to manipulate quantifiers.

The new rules allow us to get instances of quantified sentences, so that we can apply the rules we learned in Chapter 1 to the instances of quantified sentences …

And then quantify instances to generate the quantified sentences we are seeking.

What this means is …

- That all of the proof stuff we learned in sentential logic still applies, including the strategies.

- So we will still …
  1. Assume the given premises
  2. Try to apply rules to generate desired conclusion resting only on given premises *(Be methodical!)*
  3. If you need to make an additional assumption, <u>LOOK AT WHAT YOU WANT TO PROVE</u> **not** at what you have already established and make the assumption according to the hints below, and return to step 2

Sentential Strategies

- ***ATOMIC SENTENCE, NEGATION, or DISJUNCTION***
  Assume the denial of what you are trying to prove (aim for RAA)
  (If you are trying to prove a disjunction using primitive rules only, also assume one of the disjuncts and use vI and RAA to conclude its negation)
- ***CONDITIONAL***
  Assume the antecedent and try to prove the consequent (for ->I)
- ***CONJUNCTION***
  Try to prove each conjunct separately (make appropriate assumptions for proving each conjunct)
- ***BICONDITIONAL***
  Try to prove each of the corresponding conditionals separately (make the appropriate assumption for each conditional

Predicate Proof Strategy additions

The general strategy for doing proofs in predicate logic incorporates the strategy for proofs in sentential logic.

1. If you have a *Universal* sentence, use Universal Elimination to derive one or more instances as needed.
2. If you have an existential formula, assume an instance (for Existential Elimination).
3. Apply rules until you get to the desired conclusion or need to make an assumption.
4. If you need to make an assumption, *look at what you are trying to prove* and use the hints above in conjunction with these below:

If what you seeks is a …

- **Universal:**

  Try to prove an instance of it (and then do Universal Instantiation -- beware of conditions!)

- Existential:

  Either assume its negation (for RAA) or try to prove an instance of it (and then do Existential Introduction)

S87) $\exists x(Gx \,\&\, {\sim}Fx), \forall x(Gx \rightarrow Hx) \vdash \exists x(Hx \,\&\, {\sim}Fx)$

1	(1)	$\exists x(Gx \,\&\, {\sim}Fx)$	A
2	(2)	$\forall x(Gx \rightarrow Hx)$	A
3	(3)	$Ga \,\&\, {\sim}Fa$	A (for $\exists$E on 1)
2	(4)	$Ga \rightarrow Ha$	2 $\forall$E
3	(5)	Ga	3 &E
2,3	(6)	Ha	4,5 $\rightarrow$E
3	(7)	${\sim}Fa$	3 &E
2,3	(8)	$Ha \,\&{\sim}Fa$	6,7 &I
2,3	(9)	$\exists x(Hx \,\&\, {\sim}Fx)$	8 $\exists$I
1,2	(10)	$\exists x(Hx \,\&\, {\sim}Fx)$	1,9 $\exists$E (3)

S88) $\exists x(Gx \,\&\, Fx), \forall x(Fx \rightarrow {\sim}Hx) \vdash \exists x{\sim}Hx$

1	(1) $\exists x(Gx \,\&\, Fx)$	A
2	(2) $\forall x(Fx \rightarrow {\sim}Hx)$	A
3	(3) $Gb \,\&\, Fb$	A (for $\exists$E on 1)
2	(4) $Fb \rightarrow {\sim}Hb$	2 $\forall$E
3	(5) Fb	3 &E
2,3	(6) ${\sim}Hb$	4,5 $\rightarrow$E
2,3	(7) $\exists x{\sim}Hx$	6 $\exists$I
1,2	(8) $\exists x{\sim}Hx$	1,7 $\exists$E (3)

S89) $\forall x(Gx \rightarrow \sim Fx), \forall x(\sim Fx \rightarrow \sim Hx) \vdash \forall x(Gx \rightarrow \sim Hx)$

1	(1) $\forall x(Gx \rightarrow \sim Fx)$	A
2	(2) $\forall x(\sim Fx \rightarrow \sim Hx)$	A
3	(3) Gc	A (for $\rightarrow$I)
1	(4) $Gc \rightarrow \sim Fc$	1 $\forall$E
2	(5) $\sim Fc \rightarrow \sim Hc$	2$\forall$E
1,3	(6) $\sim Fc$	3,4 $\rightarrow$E
1,2,3	(7) $\sim Hc$	5,6 $\rightarrow$E
1,2	(8) $Gc \rightarrow \sim Hc$	7 $\rightarrow$I (3)
1,2	(9) $\forall x(Gx \rightarrow \sim Hx)$	8 $\forall$I

Derived Rule: Quantifier Exchange

- As with derived rules in chapter 1, you don’t need this rule, as you can accomplish the inference using the primitive rules alone, but it can be helpful with longer proofs:

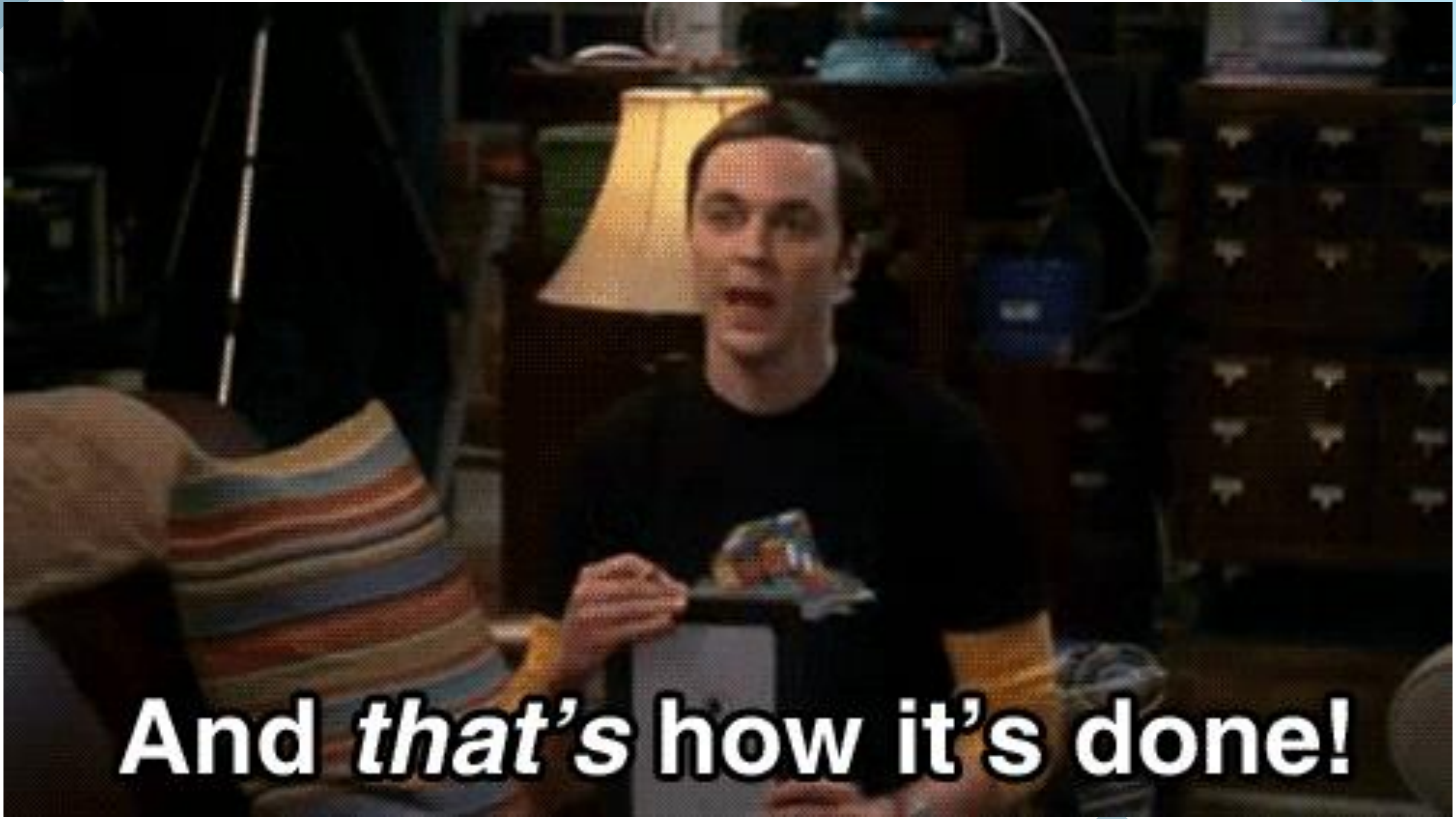
$$\sim\forall xFx \ \ \text{-||-}\ \exists x\sim Fx$$

$$\sim\exists xFx \ \ \text{-||-}\ \forall x\sim Fx$$

- You may see this referenced in the solutions provided by the text and the logic machine, annotated as QE

$\forall x(Fx \rightarrow \sim Gx) \vdash \sim\exists x(Fx \,\&\, Gx)$

1	(1)	$\forall x(Fx \rightarrow \sim Gx)$	A
2	(2)	$\exists x(Fx \,\&\, Gx)$	A (for RAA)
3	(3)	$Fa \,\&\, Ga$	A (for $\exists$E on 2)
1	(4)	$Fa \rightarrow \sim Ga$	1 $\forall$E
3	(5)	Fa	3 &E
1,3	(6)	$\sim Ga$	4,5 $\rightarrow$E
3	(7)	Ga	3 &E
1,3	(8)	$\sim \exists x(Fx \,\&\, Gx)$	6,7 RAA (2)
1,2	(9)	$\sim \exists x(Fx \,\&\, Gx)$	2,8 $\exists$E (3)
1	(10)	$\sim \exists x(Fx \,\&\, Gx)$	2,9 RAA (2)


And *that's* how it's done!